Ren Zhuang

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Seeking a Ph.D. environment that nurtures independent exploration, rigorous feedback, sustained intellectual exchange, and open-ended inquiry.

Research Interests: Large Language Model Reasoning, Reinforcement Learning, Test-Time Inference, Autonomous Agents, Cognitive Modeling, Interpretability and Alignment

EDUCATION

Hangzhou Normal University Sep 2022 – Jan 2025
M.Eng. in Software Engineering, with a focus on Natural Language Processing *Hangzhou, China*

- Average score: 86.64/100
- Key coursework: Deep Learning, Advanced Computer Graphics, Advanced Data Structures and Algorithms, Computer Networks, Database Systems

Shaoxing University Sep 2016 – Jul 2020
B.Eng. in Mechatronic Engineering, spanning Control & Mechanical Systems *Shaoxing, China*

- Average score: 72.75/100. After shifting to AI, academic engagement was strengthened.
- Relevant coursework: Calculus, Linear Algebra, Probability and Statistics, Programming in C, Microcomputer Project Laboratory, Feedback Control

PUBLICATIONS

[1] **Zhuang, R.**, Wang, B., and Sun, S. "Lifted State Policy Optimization: Internalizing Self-Correction through Transition Geometry." In *Proceedings of the 40th Annual Conference on Neural Information Processing Systems (NeurIPS 2026)*. [arXiv].

[2] **Zhuang, R.**, Wang, B., and Sun, S. "The Geometric Reasoner: Manifold-Informed Latent Foresight Search for Long-Context Reasoning." In *Proceedings of the 43rd International Conference on Machine Learning (ICML 2026)*. [arXiv].

[3] **Zhuang, R.** "Adaptive GoGI-Skip: Coupling Goal-Gradient Importance with Dynamic Uncertainty for Efficient Reasoning." In *Proceedings of the 35th International Joint Conference on Artificial Intelligence (IJCAI 2026)*. Oral. [DOI]. [arXiv].

[4] **Zhuang, R.** "The Ratchet Effect in Silico: How Interaction Drives Cumulative Intelligence in Large Language Models." In *Proceedings of the 48th Annual Meeting of the Cognitive Science Society (CogSci 2026)*. [eScholarship]. [arXiv].

[5] **Zhuang, R.**, Wang, B., Sun, S., and Lu, K. "Factored Internal Verification Enhances Large Language Model Factuality." In *Proceedings of the 32nd International Conference on Neural Information Processing (ICONIP 2025)*. [DOI].

[6] **Zhuang, R.**, Wang, B., and Sun, S. "Twin Prompt: An End-to-End Framework Inspired by Human Cognition for Navigating Language Model Reasoning." In *Proceedings of the 2025 IEEE International Conference on Systems, Man, and Cybernetics (SMC)*. Oral. [DOI].

[7] **Zhuang, R.**, Wang, B., Sun, S., Wang, Y., Ding, Z., and Liu, W. "Unlocking Chain of Thought in Base Language Models by Heuristic Instruction." In *Proceedings of the 2024 International Joint Conference on Neural Networks (IJCNN)*. [DOI].

RESEARCH EXPERIENCE

Efficient Inference and Geometric Reasoning Jan 2025 – Present
Research Assistant, Hangzhou Normal University *Fuzhou, China (Remote)*

- Developed LSPO, a lifted-state policy optimization framework that assigns credit to local self-correction transitions and internalizes revision without an inference-time loop, improving direct pass@1 on Qwen3.5-27B from 65.9% to 71.8% across six benchmarks, within 0.1 points of two-pass reflection at lower latency.

- Initiated and led TGR, a manifold-informed latent foresight search framework for long-context reasoning, improving pass@k AUC by up to 13 percentage points with only 1.1–1.3$\times$ overhead.
- Designed and implemented Adaptive GoGI-Skip to reduce redundant chain-of-thought steps, cutting CoT token volume by $>$45% and accelerating inference up to 2.0$\times$ while keeping accuracy close to the baseline.

Self-Correction and Cognitive Scaling in LLM Reasoning Sep 2023 – Jan 2025
Hangzhou Normal University, Zhejiang Wuxin Technology Co., Ltd. *Hangzhou, China*

- Led the master's thesis project on an LLM-powered climate-change dialogue system, training C^{2}Embedding and C^{2}Reranker from pretrained base models and developing the system with domain RAG and Twin Prompt.
- Proposed Twin Prompt, a cognitive-inspired prompting method that paraphrases both queries and answers, improving response matching by 19.4% over zero-shot CoT across nine LLMs.
- Conducted independent research on FactPrompt, a factored internal verification method for LLM factuality, improving factual accuracy by 4.3 percentage points on average and up to 8.37 points across 10 datasets.
- Introduced POLIS, a multi-agent ratchet framework for cognitive scaling, yielding gains of 8.8–18.9 percentage points for 1–4B-parameter model populations.

Structured Reasoning in Base Language Models Nov 2022 – May 2023
Zhejiang-Cyprus Joint Laboratory of Smart City and Mobile Health *Hangzhou, China*

- Formulated Heuristic CoT and SPIRE for structured reasoning in base language models, achieving +4.12% F1 over zero-shot CoT on SQuAD v1.1 and +17.47% F1 for ALBERT-base in ablation.

PROJECTS

Digital Twin Application Jul 2024 – Sep 2024
China Industrial Cybersecurity Center (CIC) *Wuhan, China*

- Assisted in creating digital-twin software prototypes for industrial-control scenarios, supporting the translation of domain requirements into interactive demonstrations.

Blockchain and Digital Solution Development Sep 2022 – Jan 2024
Yangtze Delta Region Institute of Tsinghua University *Hangzhou, China*

- Participated in applied blockchain and digital solution projects, contributing to requirements analysis, system testing, and technical documentation.
- Supported digital solution development for enterprise-facing scenarios, helping align stakeholder requirements with software implementations.

HONORS AND AWARDS

Awards

- Outstanding Master's Thesis Award, 2025
- Outstanding Graduate Student in Professional Practice, 2024

Scholarships

- 2nd-class Scholarship for Postgraduate Studies, 2024
- 3rd-class Scholarship for Postgraduate Studies, 2023
- 1st-class Scholarship for Postgraduate Studies, 2022

SKILLS

- **Languages**: Mandarin (native), English (IELTS Academic 7.0 overall; professional working proficiency).
- **Programming Languages**: Python, C/C++, Rust.
- **Engineering**: PyTorch, CUDA, Transformers, LlamaIndex, prompt engineering, SFT, RL training, multi-GPU/node deployment, agent workflows.
- **Systems and Tools**: Linux, Docker, Bash, Git, LaTeX.